

ECSE 2500

Lee 4

Jan. 23

Topic: Applications of Axioms of probability and conditional probability

□ Sanity check

The events $\{X=1\}$, $\{X=2\}$, $\dots$, $\{X=n\}$, $\dots$ are all disjoint, and therefore from Axiom 3, we have

$$P(S) = \sum_{n=1}^{\infty} P(X=n)$$

$$= \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^{n-1} \left(\frac{1}{6}\right)$$

$$= \frac{1}{6} \times \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^{n-1}$$

$$= \frac{1}{6} \times \frac{1}{1 - \frac{5}{6}} = \frac{1}{6} \times 6$$

$$= 1 \quad \leftarrow \text{Satisfy Axiom 2}$$

Here we used the important formula for the sum of a Geometric Series

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \text{ given } 0 < a < 1.$$

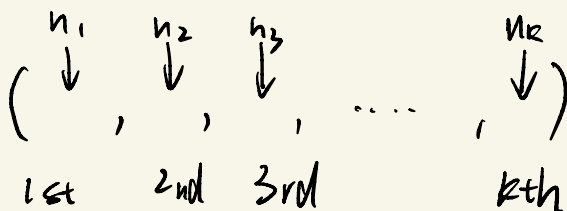
These kinds of counting problems can be approached with ideas from Combinatorics.

Key things to know:

1. K-tuples. Problems where there are n_1 choices for the first variable, n_k choices for the k th variable. Then there are in total

$$n_1 \times n_2 \times \dots \times n_k = \prod_{i=1}^k n_i$$

distinct ordered k -tuples.



Example

License plate: 3 letters followed by 4 digits. How many possible license plates?

$26^3 \times 10^4$ in total
license plates

DMV New York State



26 possibilities 26 26 10 10 10 10 possibilities

Why this is related to probability?

Q: What is the probability a randomly selected license plate is of the form

A _ _ _ 7 _ _ ?

We can tackle this problem by computing

$$\begin{aligned}\text{Prob} &= \frac{\text{number of plates having } Axxx7xx}{\text{total number of license plates}} \\ &= \frac{26^2 \times 10^3}{26^3 \times 10^4} \\ &= \frac{1}{260}\end{aligned}$$

2. Ordered samples with replacement.

Choose k objects from a set of n with replacement, i.e., pick objects, observe its

properties, put it back (so it can be selected again next time). Then applying (1) there are n^k ordered k -tuples.

$(Ob_1, Ob_2, \dots, Ob_k)$
 $\uparrow \quad \uparrow \quad \quad \quad \uparrow$
 $n \text{ choices } n \quad \quad \quad n \text{ choices}$

Example

We choose a password by randomly selecting 4 8-bit ASCII characters.

How many passwords are there?

(Pw_1, P_2, P_3, P_4)
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $2^8 \text{ choices } 2^8 \quad 2^8 \quad 2^8 = 256$
 8 bits

$(256)^4$ total number of passwords.

3. Ordered samples without replacements

Choose k objects from a set of n objects.
But object cannot be chosen again after it is picked. So # of possible objects decreases by 1 each time we select an object.

By Model (1), there are
ordered samples without replacements

$$\begin{aligned} & n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) \\ &= \frac{n!}{(n-k)!} = \frac{n \cdot (n-1) \cdot \cdots \cdot 1}{\cancel{(n-k)} \cdot \cancel{(n-k-1)} \cdot \cdots \cdot 1} \end{aligned}$$

Example

We choose a password by randomly selecting 4 8-bit ASCII characters such that no characters can be used twice.
How many possible passwords?

$$(256) \cdot (255) \cdot (254) \cdot (253)$$

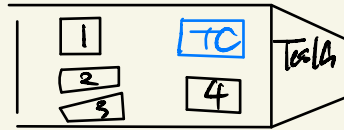
4. Permutations of objects

How many ways are there to arrange n distinct objects? Like the previous problem with $k=n$ (i.e., keep drawing from n objects without replacements until no more options).

There are $n!$ permutations of n distinct objects.

Example

A car has 4 seats. How many ways are there for 4 people to sit in this car?



$4 \times 3 \times 2 \times 1$ arrangements

5. Sampling without replacement or ordering

This is for problems where we **don't care** **about the order** of the objects and is typically posed as

"How many ways are there to choose k out of n objects?"

Without ordering, "1 2 3" and "1 3 2" are same
We can use the notation

C_k^n to denote "n choose k"

$$C_k^n = \frac{n!}{(n-k)! k!} \quad \text{much smaller than } \frac{n!}{(n-k)!}$$

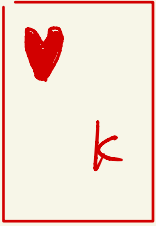
↑
Ordered samples

Example

How many 3-person groups can be formed from a pool of 10 candidates?

$$C_3^{10} = \frac{10!}{7! 3!} = \frac{10 \times 9 \times 8}{3 \times 2} = 120$$

Example



13 cards per color

Choose 5 cards at random from the deck of 52 cards. What is the probability that exactly 2 are Hearts?

There are C_5^{52} ways to choose 5 cards out of 52 cards.

There are C_2^{13} ways of selecting 2 cards from all 13 heart cards.

There are C_3^{39} ways of selecting 3 non-heart cards.

Therefore, the probability

$$= \frac{C_2^{13} C_3^{39}}{C_5^{52}} P(\text{exactly 2 cards out of 5 cards are heart})$$

- Assign probabilities to events in a continuous sample space (Next)

So far we could enumerate and assign a probability to each discrete outcome.

When we have a continuous sample space, we cannot assign a probability to an exact outcome; so we must assign a probability to a set of outcomes.

Example: Randomly pick a ^{continuous} number between 0 and 1.

What is the probability $P(\frac{1}{2})$?

It must be 0! It seems counterintuitive at first glance. But think about it. Suppose each real number was equally likely to be selected. Since there are infinitely many numbers in $[0, 1]$, the sum of probabilities will be infinite!

This violates Axiom 2.